Lecture 8

Joint Distributions of Continuous Random Variables

13.04.2026

Joint distribution of two continuous random variables

Definition

Random variables X and Y are jointly continuous if there exists a function $f(x, y)$ such that

1. *Non-negative $f(x, y) \geq 0$, for any $x, y \in \mathbb{R}$, and*
2. $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1.$

$f_{X,Y}(x, y)$ is called the joint probability density function of X and Y.

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- Connection between joint pdf and joint cdf

$$F(a,b) = P(X \leq a, Y \leq b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f(x,y)\, dx\, dy$$

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$$f(x,y) = \frac{\partial^2}{\partial x \partial y} F(x,y)$$

Marginal pdfs

Marginal probability density functions are defined in terms of "integrating out" one of the random variables.

$$f_X(x) = \int_{-\infty}^{\infty} f(x,y)\ dy$$

$$f_Y(y) = \int_{-\infty}^{\infty} f(x,y)\ dx$$

Example

The joint pdf of a bivariate random variables (X, Y) is given by

$$f(x,y) = \begin{cases} k(x+y), & 0 < x < 2, 0 < y < 2, \\ 0, & \text{otherwise,} \end{cases}$$

where k is a constant.

a) Find the value of k.
b) Find the marginal pdf's of X and Y.
c) $P(X+Y \leq 1) =?$
d) $P(X+Y \leq 3) =?$

Example

The joint pdf of a bivariate random variables (X, Y) is given by

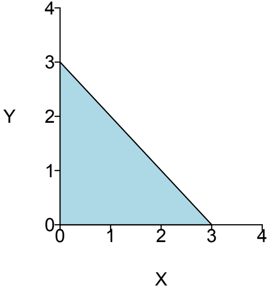
$$f(x,y) = \begin{cases} cx^2y, & x^2 \leq y \leq 1, \\ 0, & \text{otherwise,} \end{cases}$$

Find:

a) c.
b) $P[X \geq Y]$.
c) $f_X(x)$ and $f_Y(y)$.

Example

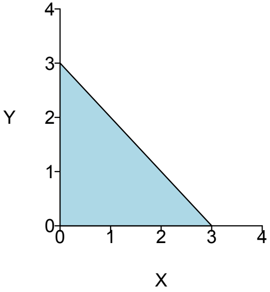
Let X and Y be drawn uniformly from the triangle below. Find the joint pdf $f_{X,Y}(x, y)$.



Let X and Y have the following joint pdf

$$f(x,y) = \begin{cases} \frac{2}{9} & \text{for } x \geq 0, y \geq 0 \text{ and } x + y \leq 3 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal pdf $f_X(x)$.



In the previous example, find $P(X < Y)$.

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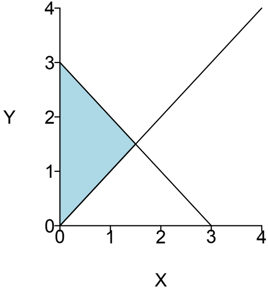

Joint distribution of two continuous random variables

- Joint pdf
  - Non-negative $f_{X,Y}(x,y) \geq 0$, for any $x, y \in \mathbb{R}$
  - $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x,y)\ dx\ dy = 1$
  - For any set $C \subset \mathbb{R}^2$,

$$P[(X,Y) \in C] = \iint_{(x,y) \in C} f_{X,Y}(x,y)\ dx\ dy$$

- Between joint cdf and joint pdf

$$F_{X,Y}(a,b) = \int_{-\infty}^{b} \int_{-\infty}^{a} f_{X,Y}(x,y)\ dx\ dy$$

$$f_{X,Y}(x,y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x,y)$$

- Marginal pdfs

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\ dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\ dx$$

Outline

Independent random variables

Definition

Random variables X *and* Y *are independent if any real sets* $A, B \subset \mathbb{R}$*,*

$$P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$$

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- If both are continuous, pdf: for any $x, y \in \mathbb{R}$

$$f_{X,Y}(x, y) = f_X(x)f_Y(y)$$

A man and a woman decide to meet at a certain location. If each of them independently arrives at a time uniformly distributed between 12 noon and 1 P.M., find the probability that the first to arrive has to wait longer than 10 minutes.

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Let random variables X, Y be the time they arrive (uniform between 0 to 60 minutes).